

# MATLAB & Simulink

Engineering maths and simulation · Equigrow short notes

## Basics

```
x = 0:0.01:2*pi;           % vector from 0 to 2π
y = sin(x);
plot(x, y); grid on; xlabel('t'); ylabel('sin t'); title('Sine wave')
A = [1 2; 3 4]; b = [5; 6];
sol = A \ b;                % solve Ax = b
A .* A                      % element-wise; A * A is the matrix product
```

## Scripts & functions

```
function y = square(x)
    y = x.^2;
end
```

## Signals

```
fs = 1000; t = 0:1/fs:1; s = sin(2*pi*50*t) + 0.5*randn(size(t));
Y = abs(fft(s)); f = (0:length(Y)-1)*fs/length(Y); plot(f, Y)
```

## Simulink

- Build models from blocks (Sources, Math, Continuous, Sinks) and connect them with signals.
- Set the solver and stop time, run, and view results in the Scope block.